

Chapter 11

Apprehensional markers in Korean

Seongha Rhee^{a,b} & Tania Kuteva^{c,d}

^aMahidol University ^bHankuk University of Foreign Studies

^cHeinrich-Heine-Universität Düsseldorf ^dSOAS, University of London

This paper addresses the diachronic development and synchronic distribution of thirteen markers of apprehension in Korean. They have been grammaticalized to varying degrees from diverse constructions, forming multiple layers in contemporary Korean. The sources involve the future marker, questions, cognitive verbs, and mode- or purpose-adverbializers, among others. The notion of apprehension is either fully semanticized in the constructions or is only pragmatically inferred from them. The role of pragmatic inference is also evident in the reanalysis of a sentence-ender as an apprehensional causal connective. In many cases of pragmatic inference, subjectification from objective to subjective meanings, and directionality from epistemic possibility to apprehension are prominent.

1 Introduction

1.1 Objectives

Korean has a number of grammatical forms that encode the speaker's apprehension, i.e., apprehensionals (or “preventive markers”; Ramstedt 1997[1939], Kim 1960, Ko 1989, Son 1994). Their primary function is to warn the addressee of potentially harmful consequences of an action, or to present reasons for the speaker's action or mental state, as exemplified in (1).

- (1) a. Connective *-lla* ‘lest’
Pelley tuleo-lla changmwun tat-ala.
insect come.in-APPR window close-IMP
‘Close the window lest insects come inside.’



- b. Connective *-lkkapwa* ‘for fear of’
Cencayng-i na-lkkapwa cam-i an o-n-ta.
 war-NOM come.out-APPR sleep-NOM not come-PRS-DECL
 ‘(I) cannot sleep for fear of a war.’

Despite the early appearance of the notion in Korean linguistics, as in Ramstedt (1997[1939]), the markers have not yet received any attention from a grammaticalization perspective. In order to fill this research gap, this paper addresses the diachronic development and synchronic distribution of the apprehensionals in Korean from a grammaticalization perspective, and discusses a number of theoretically important issues that arise from the analysis.

Here we are using the term *apprehensional* for the overall domain of both bi-clausal and mono-clausal structures which in some way involve apprehension. Notice that while most of the markers we discuss entail undesirability, others can be used in more neutral contexts. As proposed in Kuteva et al. (2019: 865), it is justifiable to treat both bi-clausal and mono-clausal structures as instantiations of the same overall – apprehensional – domain, all the more so because there exists a fairly regular pathway between the two in the process of “insubordination” (Evans 2007, Evans & Watanabe 2016) leading from the bi-clausal to the mono-clausal structure.

Since Korean has a large inventory of apprehensionals, a microscopic approach to a few select forms may prove fruitful, but we adopt a more macroscopic perspective in order to describe the system of Korean apprehensionals as a whole, and thus include both weakly and strongly grammaticalized forms. We leave more fine-grained research on individual forms for future research. This paper is organized in the following manner. After a short introduction to the data sources as well as demographic, syntactic, and typological characteristics of the Korean language, in §1, we describe the grammatical means of signaling apprehension as belonging to three major categories according to the types of lexical and grammatical sources, in §2. In §3, we discuss a few theoretical issues, such as grammaticalization parameters, layering and specialization, roles of source concepts, and a few factors involved in semasiological changes such as pragmatic inference, reinterpretation, and subjectification. We summarize the discussion and conclude the paper in §4.

1.2 Data sources

The data for analysis are taken from various sources, including the Sejong Historical Corpus for diachronic investigation and the Drama and Cinema Corpus for investigation of contemporary usage.

The Sejong Historical Corpus is a 15 million-word, historical section of the Sejong Corpus, a 200 million-word corpus developed as part of the 21st Century Sejong Project by the Korean Ministry of Culture and Tourism and the National Institute of Korean Language (1998–2006). The texts in the historical section, dating from 1446 through 1923, are of diverse genres including Buddhist scriptures and commentaries, medical advisories, poems, literature commentary, books on ethics, novels, short stories, interpreter training textbooks, reports to kings, biblical tracts, newspaper articles, diaries, personal letters, etc. Of particular importance is the inclusion of 43 pieces of *sinsosel* ‘new novels’ (mostly between 1906–1913), a literary genre that bridges the classical novels and modern novels in the history of Korean literature. The genre was influenced by Western literary styles including the use of colloquial language. Since grammaticalization processes can be more easily captured in colloquial data, *sinsosel* is often the data source where emerging grammatical functions are first attested.

The data for Modern Korean is taken from the Drama and Movies Corpus. The corpus is a 24 million-word contemporary corpus, a collection of 7,454 scenarios of dramas and movies dating from 1992 through 2015, compiled by Min Li of Seoul National University. To search and process the data, the Korean concordance program UNICONC, developed by Jinho Park, was used.¹

1.3 Preliminaries on Korean

Korean is spoken by about 77.2 million people (*Ethnologue* Simons & Fennig 2018) in and around the Korean peninsula: South Korea, North Korea, China, Japan, and other Korean communities around the world. It is a head-final, agglutinating language with a SOV word order. Sentential arguments are not only omissible but such omission is often preferred when their referents are contextually recoverable. The sentence-final word is the main-clause verb with a number of morphological markers. These sentence-final verbal suffixes, collectively called “sentence-final particles” by most Korean linguists, consist of the T(ense)-A(spect)-M(odality)-Mood morphology, including such subjective and intersubjective markers as evidentiality, epistemicity, politeness, deferentiality, and honorification. These occur in ordered slots, the last of which is occupied by a sentence-ender designating the speech-act type and politeness/deferentiality.

The structure that is of particular interest for the present research is clause-embedding patterns. Clause embedding may involve direct or indirect quotation

¹We thank the developers of the program and corpora for generously granting their use for research.

(as shown in (2) and (3), respectively). The distinction between direct and indirect quotation is often difficult because the distinction is not strictly binary. A good clue in Korean is whether the embedded clause ends with a regular sentence-ender (direct speech), such as *-ta* in (2b), or with a complementizer (indirect speech), such as *-tako* in (3b). Since Korean has a large number of regular sentence-enders (Kim 2001: 147–151 lists 381 forms), but only four to five complementizers, the distinction is relatively easy. There are rare occasions, however, when the final form is unclear as to its sentence-ender or complementizer status in case the phonetically reduced complementizer, formally identical with the sentence-ender, is used. In such cases other clues may help disambiguate its status.

(2) Direct Speech

- a. (SUBJECT) “[(SUBJECT) (OBJECT) VERB]” VERB
 (NP-NOM) [(NP-NOM) (NP-ACC) V-A-T-M-END] V-A-T-M-END
- b. *Ku-ka* “[*atul-i pap-ul mek-koiss-ess-napo-ta*]”
 he-NOM [son-NOM food-ACC eat-PROG-PST-CONJ-DECL]
 chwuchukha-koiss-ess-ta.
 guess-PROG-PST-DECL
 ‘He was guessing, (saying) “(My) son seemed to be eating food (then).”

(3) Indirect Speech

- a. (SUBJECT) [(SUBJECT) (OBJECT) VERB] VERB
 (NP-NOM [(NP-NOM) (NP-ACC) V-A-T-M-COMP] V-A-T-M-END
- b. *Ku-ka* “[*atul-i pap-ul mek-koiss-ess-napo-tako*]”
 he-NOM [son-NOM food-ACC eat-PROG-PST-CONJ-COMP]
 chwuchukha-koiss-ess-ta.
 guess-PROG-PST-DECL
 ‘He was guessing that (his) son might have been eating food (then).’

A significant feature of Korean grammatical markers in general is that they signal not only garden-variety grammatical notions, such as grammatical case, local case, coordination, subordination, etc., but also diverse stance-related notions, such as tepidity, unexpectedness, beneficiality, extremity, unrealistic hypotheticality, inordinateness, etc., which are often difficult to label. Thus, Korean grammatical markers are paradigm examples of “elaborateness” or “semantically elaborate categories” (Kuteva 2009, Kuteva et al. 2019) or markers of languages

of an “esoteric” nature (Thurston 1988, Grace 1997, Crowley 2000, Wray & Grace 2007, as cited in Rice 2017, 2019).

Another important fact is that intersubjectivity marking is obligatory in Korean. Most prominently, Korean sentences, properly ended, should be marked with the speech level (*hwakyey* in Korean linguistics), depending on the relative social hierarchy between the speaker, hearer, and sentential subject. This system is complex and variable, with four to seven different levels of honorification. The rules pervasively apply to not only verbal morphologies but also nominal morphologies.

2 Grammaticalization of apprehensionals

2.1 Apprehensionals in contemporary Korean

As indicated earlier, Korean has a large number of apprehensionals with differential degrees of semanticization of the notion of apprehension and with different levels of currency. They have the status of a clausal connective (CONN), usually specifying the reason for the fear in the clause that hosts the apprehensional, or of a sentence-ender (END), simply to indicate the speaker’s emotional state of apprehension with respect to the content of the proposition. When apprehensionals occur as connectives, the clauses hosted by them will function as subordinate clauses. Some apprehensionals can be used either as a connective on the subordinate clause or as a sentence-ender on the main clause, whereas some others are used only as one or the other (cf. Ramstedt 1997[1939], Kim 1960, Ko 1989, Son 1994). The forms are listed in Table 1 with their source constructions and grammatical classes.

As shown in Table 1, the forms that end with an *-e/a* (note that these two are phonologically controlled allomorphs) can function as a clausal connective or a sentence-ender due to the fact that *-e/a* are ‘heteronyms’ (Lichtenberk 1991), carrying dual functions in the two different grammatical categories.² As a connective it marks sequentiality or causality relations between the two events denoted

²The connective function is older and its use as a sentence-ender is a 19th century innovation (Suh 1993: 139–141; see, however, Park 1996 for a different view). At the time of innovation, when a sentence ended with this form (formerly a connective), it was felt that the speaker stopped an utterance without completing it. Thus, the speech style using this sentence-ender has been named *panmal* ‘half-talk’. As a sentence-ender, *-a/e* lacks honorification or politeness, but is uniformly used across all speech acts, e.g., declarative, interrogative, imperative, hortative, promissive, etc.

Table 1: Apprehensional forms, source constructions, and grammatical categories

Form	Source construction	Gloss for source construction	Class
a. <i>-lkka</i>	< <i>-l-kka</i>	FUT-Q	CONN
b. <i>-lkkapwa</i>	< <i>-l-kka-po-a</i>	FUT-Q-see-CONN/END	CONN/END
c. <i>-lkkamwusewe</i>	< <i>-l-kka-mwusep-e</i>	FUT-Q-be.fearful-CONN/END	CONN/END
d. <i>-lkkamolla</i>	< <i>-l-kka-molu-a</i>	FUT-Q-not.know-CONN/END	CONN/END
e. <i>-lkkasiphe</i>	< <i>-l-kka-siph-e</i>	FUT-Q-think-CONN/END	CONN/END
f. <i>-lla</i>	< <i>-l-la</i>	FUT-END	END/(CONN)
g. <i>-lseyla</i>	< <i>-l-sa-i-la</i>	FUT-occasion-be-END	END/(CONN)
h. <i>-cimolla</i>	< <i>-ci-molu-a</i>	NOMZ-not.know-CONN/END	CONN/END
i. <i>-cimoluni</i>	< <i>-ci-molu-ni</i>	NOMZ-not.know-CONN	CONN
j. <i>-cimosakey</i>	< <i>-ci-mos-ha-key</i>	NOMZ-cannot-do-MODE	CONN
k. <i>-cimoshatolok</i>	< <i>-ci-mos-ha-tolok</i>	NOMZ-cannot-do-PURP	CONN
l. <i>-cianhkey</i>	< <i>-ci-ani-ha-key</i>	NOMZ-not-do-MODE	CONN
m. <i>-cianhtolok</i>	< <i>-ci-ani-ha-tolok</i>	NOMZ-not-do-PURP	CONN

by the two clauses (cf. “consolidating connective”; Koo 1987), and as a sentence-ender it marks the end of a sentence at the “intimate” speech level. It is noteworthy, in all cases except for (a) *-lkka*, (f) *-lla* and (g) *-lseyla*, that the grammatical function of the final element of the source construction determines the grammatical category of the grammaticalized form, a point observed in Rhee (2007) (see §3.4 for discussion of the exceptional cases). Note that the markers of MODE and PURP(ose) in (j) through (m) are adverbializing connectives used to form converbs (Ramstedt 1903, 1997[1939], Haspelmath 1995), called *pwutongsa* among Korean linguists.³

Another aspect evident from the list is that there is a certain, limited number of concepts that are involved in the source constructions, i.e. grammatical markers of FUT(urity), Q(uestion), MODE, and PURP(ose) and lexical verbs of perception, emotion, and cognition such as ‘see’, ‘be fearful of’, ‘not know’, and ‘think’. The majority of the apprehensionals, seven out of thirteen, developed from constructions involving future tense marker *-l* (and its allomorph *-ul*).

³An anonymous reviewer notes that these mode- and purpose-marked forms (j) through (m) exhibit differences from the rest in that they all involve a converb and a negative morpheme. It is true in that light that the paradigm of Korean apprehensionals exhibits a low level of “paradigmaticity” (Lehmann 2015[1982]), in that there are many members with dissimilar formal properties within a category. The category of apprehensionals is a functionally motivated paradigm.

Among these seven apprehensionals, five also involve a question marker. Six of them involve the nominalizer *-ci*. Similarly, six of them involve perception-emotion-cognition verbs. For the following more detailed discussion of these apprehensionals, the thirteen apprehensionals are classified into three groups according to their prominent source characteristics, i.e. those developed from constructions involving future and question markers (§2.2), those involving future markers without question markers (§2.3), and those involving the nominalizer (§2.4). It is evident that the three groups do not form mutually-exclusive classes, and thus inclusion of an apprehensional in a particular class is largely a matter of convenience. Likewise, those involving perception, emotion, and cognition verbs are not separately addressed, but are discussed in the three categories, when they are relevant.

2.2 Apprehensionals from future + question

The first group of apprehensionals are those originating from the source constructions involving the future tense marker *-l-* and the question marker *-kka*, i.e. *-lkka*, *-lkkapwa*, *-lkkamwusewe*, *-lkkamolla*, and *-lkkasiphe*. The primary function of the grammatical morpheme *-l-* is marking the future tense, indicating probability, intent, prediction, etc. but it can also indicate counterfactual or conjecture when used with a past tense marker. It is etymologically related to the prospective-marking adnominal *-l*.⁴ All these functions have the irrealis meaning in common.

2.2.1 *-lkka*

Since the apprehensional *-lkka* originated from the source construction of the future *-l-* and the interrogative sentence-ender *-kka*, the source meaning is ‘Will x ...?’, where x is a proposition, the content of which is the event or state being feared by the speaker or the sentential subject. This is the oldest apprehensional attested in the earliest *Hankul* literature (15th c., Late Middle Korean), as shown in (4):⁵

⁴Korean has different adnominal forms depending on the temporal relation of the event with respect to its referenced event (or speech time), i.e., *-(u)l* for prospective (later) as in *mek-ul pizza* ‘pizza to eat’, *-(u)n* for anterior (previous), as in *mek-un pizza* ‘pizza eaten’, and *-nun* for simultaneous (same), as in *mek-nun pizza* ‘pizza being eaten’. As indicated, the prospective adnominal *-(u)l* is conceptually and historically related to the future tense marker *-(u)l(i)-*.

⁵*Hankul*, also often written as *hangeul*, is the Korean alphabet, invented in 1443 by King Sejong the Great. Prior to its invention, Chinese characters were used for their meaning (semantogram) or sound (phonogram), thus the interpretation of some historical documents dating from the pre-*hankul* period remains controversial, despite notable advances made in recent studies.

- (4) *Swuthahoychen-i neki-toy* [...] *swutal-i wen-ul mot*
 (name)-NOM think-CONN (name)-NOM wish-ACC cannot
ilwu-lkka ha-ya [...] realize-APPR do-and
 ‘As the divinity Sutamwechon (Suddhavasehi Devehi) thought, [he came to the earth and intervened ...] lest Sudal fail to accomplish his dreams, ...’
 (1447 *Sekposangcel* 6:25a)

The apprehensional meaning associated with *-lkka*, translated as ‘lest’ in the example, largely comes from the context, as it appears that the notion of apprehension was not yet fully grammaticalized at the time. In other words, Suddhavasehi Devehi’s intervention is interpreted as a result of his fear that Sudal might fail to accomplish his dream of spiritual enlightenment. Since it is not likely that Suddhavasehi Devehi actually uttered the question ‘Will he not ...?’, the form *-lkka* on the embedded clause, while appearing to involve the question marker *-kka*, is interpreted instead as a marker of fear or serious concern (see §3.4 for more discussion on *-lkka*). Indeed, the collocation pattern in the 15th century Korean data strongly supports the hypothesis that the apprehensional meaning was pragmatically inferred from the co-occurring psych-verbs in the main clause. Among the common collocates, in addition to the light verb *ha-* ‘do/say’ in the example above, are *neki-* ‘think/regard/consider’, *pwunpyelha-* ‘distinguish’, *silumha-* ‘worry’, *ceh-/ceph-* ‘fear’, *uysimha-* ‘doubt’, *hyey-* ‘consider, calculate’, *sulh-* ‘lament’, *twuli-* ‘fear’, *kongha-* ‘fear’, *sip-/siph-* ‘think’, *minmangha-* ‘feel ashamed’, *kunsimha-* ‘worry’, *twulyeha-* ‘fear’, *sayngkakha-* ‘think’, *nyemnyeha-* ‘worry’, *acheha-* ‘dislike’, *mwusep-* ‘be afraid’, *kepulnay-* ‘be afraid’, *cAkepha-* ‘be afraid’, *kongphoha-* ‘fear’, *cosimha-* ‘be mindful’, *samkaha-* ‘be wary’, etc. Evidently, most of the co-occurring psych-verbs have a negative meaning. Thus, a sentence with an embedded question with the structure of, for instance, [‘I think [Will a war break out?’]] becomes an apprehensional sentence [‘I fear that a war might break out’]. Similarly, [‘as I think [Will a war break out?’]] becomes [‘lest a war break out’] (see §3.4 for more discussion on structural reanalysis).

Since the undesirability-of-event interpretation of *-lkka* heavily depended on the meaning of the co-occurring verb, *-lkka* sometimes occurred as a non-apprehensional marker, in neutral or even in positive (desirable) contexts, e.g. with *pala-* ‘hope’. In non-apprehensional usage, *-lkka* carries diverse modal meanings, e.g., uncertainty, possibility, conjecture, etc. In our Present-Day Korean (PDK) data, all connective usage tokens (about 991 tokens) are instances of fully developed apprehensionals.⁶ Even when it is used as the sentence-ender of

⁶The statistics given here are either actual hit-records or projections based on random sampling of 10 percent of all hits, or 500 tokens.

an embedded direct quotation and thus is immediately followed by a verb that specifies the nature of the embedded question, *-lkka* is predominantly used in contexts involving undesirable events. For instance, in such cases, 69.6% of the instances (about 2,454 tokens) involve a verb with meanings related to worries and fears, while 30.4% of them (about 1,072 tokens) appear in neutral contexts (e.g., *ha-* ‘say’, *sayngkakha-* ‘think’), and no instances are found in positive contexts. This suggests a scenario where, as *-lkka* becomes used in increasing frequency in contexts with verbs with negative connotation, it has developed into an apprehensional connective even without an accompanying verb of undesirable meaning (see §3.3 for more discussion on semantic absorption). In PDK, the apprehensional *-lkka* is moderately frequent but is yielding the functional primacy to its reinforced innovation *-lkkapwa* discussed immediately below (see §3.2 for the synchronic states of affairs regarding the different apprehensional forms).

2.2.2 *-lkkapwa*

The source construction of *-lkkapwa* is *-l-kka-po-a*, which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *po-* ‘see’ and the sentence-ender/connective *-a*, thus yielding the compositional meaning of ‘(I) see, (saying) “Will x ...?”’ (when *-a* is a sentence-ender) or ‘as (I) see, (saying) “Will x ...?”, ...’ (when *-a* is a connective).⁷ From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. ‘(I) fear x might happen’ or ‘as (I) fear x might happen’ or ‘lest x should happen.’

The apprehensional *-lkkapwa* is first attested in the late 19th century data, i.e. the Modern Korean period. Among the earliest attestations is (5), in which *-lkapoa*, the historical form of *-lkkapwa*, occurs.

- (5) *Calmosha* *tAka os-ul* *tuley-kena* *phAymwul-ul sangha-lkapoa*
 by.accident dress-ACC make.dirty-or jewelry-ACC damage-APPR
cosim-i *toy-ya* ...
 concern-NOM become-and
 ‘As (I) became worried lest my (rented silk) dress become sordid or my
 (rented) jewellery become scratched [at the party], (I hurried home to
 take them off and return them to you).’ (1911 *Welhakain* 23)

⁷As an anonymous reviewer notes, the verb *po-* ‘see’ has a large number of semantic designations and has developed into diverse grammatical markers (see Rhee 2019, 2020, 2021a,b among many others, for discussion of semantic extension and grammaticalization of the verb).

Even though it has a short history, the apprehensional connective *-lkkapwa* is the most frequently used apprehensional in PDK. Unlike in the earlier form *-lkka*, from which it originated as a reinforced form, the apprehensional meaning is fully semanticized in the connective *-lkkapwa*, i.e., the marker is not used in neutral or positive contexts (e.g., occurring with such verbs as *ha-* ‘say’ (the historical descendant of *ha-*), *sayngkakha-* ‘think’, *pala-* ‘hope’, *kotayha-* ‘look forward to’, etc.). It is found in a corpus search that all occurrences of the connective *-lkkapwa* are of the apprehensional usage (about 6,697 tokens). In PDK it can also be used as the apprehensional sentence-ender, through a process of insubordination, as exemplified in the constructed examples in (6).⁸

- (6) a. APPR connective
Cencayng-i na-lkkapwa tteli-n-ta.
 war-NOM come.out-APPR shiver-PRS-DEC
 ‘I am shivering for fear that a war might break out.’
- b. APPR sentence-ender
Cencayng-i na-lkkapwa.
 war-NOM come.out-APPR
 ‘A war might break out (I’m afraid).’

2.2.3 *-lkkamwusewe*

The source construction of *-lkkamwusewe* is *-l-kka-mwusep-e*, which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *mwusep-* ‘be fearful’ and the sentence-ender/connective *-e*, thus yielding the compositional meaning of ‘(I) am fearful, (saying) “Will x ...?”’ (when *-e* is a sentence-ender) or ‘as (I) am fearful, (saying) “Will x ...?”, ...’ (when *-e* is a connective). From these meanings of the source construction, the apprehensional interpretations emerge, e.g. ‘(I) fear x might happen’ or ‘as (I) fear x might happen’ or ‘lest x should happen.’

Historically, the forerunner construction of the apprehensional *-lkkamwusewe* is first attested in the 19th century data in the form of *-lkkamwusewese*, containing

⁸In this process, however, there also occurred a functional split, i.e., depending on the context, it signals apprehension (e.g., ‘I fear that ...’), the speaker’s tentative intention (e.g., ‘I might as well ...’), or inference (e.g., ‘It seems that ...’). As a connective it only marks apprehension. As a sentence-ender, it can, in principle, have any of these three functions, but all sentence-ender instances found in the corpus are of the tentative-intentional function (i.e., non-apprehensional), a state of affairs indicating that the divergence from its apprehensional source is nearly complete in the domain of sentence-enders in PDK.

the final particle *-esye*, a tonic variant of *-e* with a fortified sequential or causal meaning, as shown in (7):

- (7) [The magistrate who appeared without notice protested why there were no toast songs in the drinking party, but ...]
... *kisA**yng-tul-un* *chAykmangha-lkamwusewesye mos*
entertainer-PL-TOP scold-APPR cannot
ha-te-n *cha-i-la.*
do-RETRO-ADN point-be-DECL
'... (in fact) the entertainers had not (sung) lest (the magistrate) might scold (them for partying).' (1896 *Toklipsinmwun*)

Example (7) describes a scene in which the magistrate who appeared without notice pretends to be unhappy about the entertainers not singing at the party thrown by the low-ranking officials. The entertainers did not sing but instead remained quiet because the magistrate was known to be stern and austere and thus refrained from any unnecessary spending of the government fund. From the context the entertainers could have been really ‘afraid’ (*mwusep*- ‘be fearful’) or they were simply ‘concerned’ (the context supports the former interpretation). The apprehensional *-lkkamwusewe* is still in use in PDK, though not occurring at a high frequency. A corpus search returns only 86 instances of *-lkkamwusewe/-lkkamwusep*, all carrying the apprehensional function either as a connective (31.4%) or as a sentence-ender (68.6%). In the PDK usage, in spite of the presence of *mwusep*- ‘be fearful’, the construction in general does not denote genuine fear but much milder concern evoked by the situation, analogous to the English preface discourse marker *I’m afraid* (*but ...*). Since the early attestations invariably encoded genuine fear (cf. 7), the PDK usage of *-lkkamwusewe* is evidently the result of semantic bleaching or generalization, which is a common concomitant, or even a prerequisite, of grammaticalization (Bybee & Pagliuca 1985, Bybee et al. 1994; see also §3.1).⁹

2.2.4 -lkkamolla

The source construction of *-lkkamolla* is *-l-kka-molu-a*, which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *molu-*

⁹We are grateful that Eva Schultze-Berndt and Marine Vuillermet (p.c. August 5, 2020) brought to our attention Jing-Schmidt & Kapatsinski (2012) reporting similar phenomena in English, Mandarin and Russian.

‘not know’ and the sentence-ender/connective *-a*, thus yielding the compositional meaning of ‘(I) do not know, (saying) “Will x ...?”’ (when *-a* is a sentence-ender) or ‘as (I) do not know, (saying) “Will x ...?”, ...’ (when *-a* is a connective). From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. ‘(I) fear x might happen’ or ‘as (I) fear x might happen’ or ‘lest x should happen.’ The question arises in the mind of the speaker because of absence of knowledge or uncertainty, which also may arouse some form of fear.

The apprehensional *-lkkamolla* is first attested in the 17th century data, i.e., the Early Modern Korean period, and is still in use, though not occurring at a high frequency, in PDK. An example among the earliest instances of its use, in the form of *-lksamolna*, is shown in (8).

- (8) [I want to get a couple of poems and some calligraphy from him, but ...]
 ... cyecye-y ei nyeki-si-lksamolna cacyehA-na-ita.
 he-NOM how think-HON-APPR restrain.self-IND-DECL
 ‘... (I) restrain myself from (requesting) fearing how he would react/think.’
 (1687 *Kwuwunmong* SNU Archive)

As *-lkkamolla* is the least frequent apprehensional, a corpus search returns only 55 hits, of which 40 are of the apprehensional usage. In non-apprehensional usage, *-lkkamolla* simply denotes absence of knowledge (as in ‘I asked him because I didn’t know if ...’). Even though the earliest attestation is of the connective usage, as exemplified in (8), all instances of *-lkkamolla* found in the corpus are of the sentence-ender function.

2.2.5 *-lkkasiphe*

The source construction of *-lkkasiphe* is *-l-kka-siph-e*, which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *siph-* ‘think, inclined to think’ and the sentence-ender/connective *-e*, thus yielding the compositional meaning of ‘(I) think, (saying) “Will x ...?”’ (when *-e* is a sentence-ender) or ‘as (I) think, (saying) “Will x ...?”, ...’ (when *-e* is a connective). From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. ‘(I) fear x might happen’ or ‘as (I) fear x might happen’ or ‘lest x should happen.’ The speaker is inclined to think or pose a question about a potential future event, which arouses some form of fear or serious concern.

The lexical verb *siph-* in the source construction has undergone a series of semantic changes whereby a number of grammatical forms emerged in history (see Kim 2010, Rhee 2021b,c for the verb’s diverse grammaticalization scenarios

including the evidentiality function). The earliest precursor forms of *-lkkasiphe* appear as *-lkosip-END* (e.g., *-lkosipeyla*) and *-lkasipe* in Late Middle Korean, but whether those forms in the examples indicated mere inquisitive thought or fear is unclear. An unequivocal example of the apprehensional use is attested in the 17th century data (Early Modern Korean) of a wartime diary written by a court lady (note that the exact year of authorship is unknown):¹⁰

- (9) *Skwum-i ha pencaphA-ni na-y cwuk-ulkasipu-te-la.*
 dream-NOM very be.disorderly-as I-NOM die-APPR-RETRO-DECL
 ‘The dreams were so disconcerting that I feared that I might die.’ (163x
Pyengcailki 410)

The semantic change from ‘think’ to ‘be afraid’ must be the result of subjectification prompted by inferences from the context of undesirable events in which the verb *siph-* occurs. For instance, Rhee (2021c) shows that the earliest meaning of the lexeme *siph-* ‘think’ has undergone a series of changes largely dependent on the co-occurring words, and developed into the markers of desiderative, epistemic-evidential modality, apprehensional, tentative intention, similitive and evaluative connective, and discourse markers (see also Kim 2010 and Rhee 2021b for discussion of the grammaticalization of the verb *siph-*). In PDK *siph-* no longer retains its original meaning ‘think’, but is present only as a part of grammatical markers including the apprehensional *-lkkasiphe*. This form is among the least frequent apprehensionals; a corpus search returns 112 instances of *-lkkasiphe*, all of which function as an apprehensional connective. The functional split between sentence-ender and connective function, discussed for *-lkkapwa* above, is also observed with *-lkkasiphe*. In other words, when it is used as a sentence-ender it marks the speaker’s tentative intention, rather than apprehension.

2.3 Apprehensionals from future

The next group of apprehensionals involving the future marker *-l-* in their source constructions are *-lla* and *-lseyla*. These two are different from those illustrated in §2.2 in that they do not involve the interrogative sentence-ender *-kka*.

¹⁰Interpretation of historical data can be variable and often debatable. However, non-apprehensional interpretations of (9), e.g., desiderative, tentative intentional, inferential, etc., are contextually inappropriate.

2.3.1 *-lla*

The source construction of *-lla* is *-l-la*, which involves the future tense marker *-l-* and the sentence-ender *-la*. The sentence-ender *-la* is no longer in common use in PDK, and its grammatical status and function are controversial. For instance, Ramstedt (1997[1939]: 85), in line with Underwood (1915[1890]), analyzes *-lla* as consisting of the future *-l-* and the vocative *-a*, labeling it as ‘preventive’, which, however, is disputed in Ko (1989: 293, 341–342), who regards the form as mono-morphemic (i.e., unanalyzable) and labels it as a marker of *kyengkyyeyp* ‘premonitive’. Indeed, the analysis of the future marker immediately followed by the vocative is in violation of morphosyntactic rules of Korean. The final element *-la* seems to be a declarative sentence-ender with an overtone of exclamation or surprise, since its formal and functional relative, the sentence-ender *-ela*, is one of the exclamative/mirative sentence-enders in PDK. With this analysis the construction *-l-la*, future-declarative, yields the compositional meaning ‘x will ...!’, which is functionally reinterpreted as ‘(I) fear x might happen.’ There is nothing that compels such reinterpretation, but such a reinterpreted meaning becomes favored, which results in the emergence of an apprehensional. Therefore, it is through reinterpretation that the mere possibility of something taking place in the future comes to signal the speaker’s fear (see §3.4 for more discussion).

A forerunner construction of *-lla* is first attested in the 17th century data, but this did not have a dedicated apprehensional meaning as it does now, but only signaled possibility and conjecture. In the 18th century, *-lla* developed into a marker of tentative intention with first-person subjects. With second- or third-person subjects, it developed into the apprehensional, which then became highly frequently used. The usage of possibility- or conjecture-marking survived until the beginning of the 20th century. In PDK, it is an unequivocal apprehensional, even with first-person subjects, but its frequency has considerably declined. The decline seems to be due to the functional specialization of its competitor *-lkkapwa*. Most attestations in the corpus are those of the homophonic marker of intention, a variant of *-llyeko*. Example (10) is one of the earliest attestations of the apprehensional *-lla*.

(10) [In preparation of a funeral]

Pang tewu-lla pwul stAy-cimal-ko ko-y tuleka-lla
 room be.hot-APPR fire make-PROH-and cat-NOM enter-APPR
kwistol mak-ela
 flue.opening block-IMP

‘Don’t make fire (for floor-heating) lest the room become too hot (for the corpse), and block the flue opening lest a cat enter (through it) [which is ominous according to superstitions].’ (ca. 1870 *Pakhungpoka*)

In (10) the grammatical status of the apprehensional *-lla* is unclear, i.e., whether it is a sentence-ender, resulting in the meaning of ‘I’m afraid the room might become too hot’, or a connective, resulting in the meaning of ‘lest the room become too hot’ as suggested in the translation. The uncertainty of grammatical status is an intriguing issue, which will be discussed in §3.4. The apprehensional *-lla* is infrequent in PDK. For instance, a corpus search returns 184 instances of the apprehensional *-lla*, 173 tokens of which (94.0%) are of the sentence-ender function and 11 tokens (6.0%) are of the connective function.

2.3.2 *-lseyla*

The source construction of *-lseyla* is *-l-sa-i-la*, which involves the future tense marker *-l-*, the defective noun *sa* ‘occasion’ functioning as a nominalizer, the copula *i-* and the declarative sentence-ender *-la*, thus yielding the compositional meaning of ‘this occasion is that x will occur!’¹¹ From this meaning of the source construction, the apprehensional interpretations emerge, e.g. ‘(I) fear x might happen’ or ‘as (I) fear x might happen.’ A simple statement of a possible future event with an exclamative overtone (see *-lla* above) gives rise to the apprehensional meaning.

The forerunner construction of *-lseyla* is frequently attested in Late Middle Korean with an overtone of exclamation and at the same time signaling a causal relation between the two connected clauses. These early examples did not carry the apprehensive function. The unambiguous apprehensional function is first attested in 1703, and became very frequent in folk songs in Early Modern Korean. Example (11) is from an 18th century collection of songs.

- (11) [Attracted by the spring breeze I stroll to the stream, only to find
countless fallen flowers in the water.]
Cye kos-a senwen-ul nam al-lseyla stenaka-cimal-wala.
that flower-VOC fairyland-ACC other know-APPR leave-PROH-IMP
‘Oh, you, flowers, don’t flow down (the stream) lest others would find out
about the fairyland (upstream).’ (ca 1767 *Haytongkayo* Cwussipon)

¹¹The source composition of the portmanteau form *-lseyla* (and its historical variant *-lsyeyla*) has been a point of controversy (see Jang 2002: 137–138). Considering the involvement of the copula *i-* and the declarative sentence-ender *-la*, we hypothesize that the preceding formant is a nominal, and thus the semantically-bleached defective noun *sa* ‘time/occasion’ in Late Middle Korean is the best candidate, both formally and semantically well-motivated. The defective noun *sa* ‘time/occasion’ originally denoted ‘the sun’ from which the notion ‘temporal/spatial expanse or interval’ was derived (Jeong 2003: 61; see also Rhee & Koo 2021).

Similar to *-lla* in (10) in the preceding discussion, the apprehensional *-lseyla* in (11) is ambiguous with respect to its grammatical status, i.e., whether it is a sentence-ender resulting in the meaning of ‘I’m afraid other people might find out about this fairyland’ or a connective resulting in the meaning of ‘lest others would find out about this fairyland’ as provided in the translation of (11) (see §3.4 for more discussion).

Even though *-lseyla* is often considered one of the representative apprehensional forms in the history of Korean, it is not frequently used anymore in PDK and carries the flavor of archaism. A corpus search shows 233 tokens of *-lseyla*, all in the apprehensional connective usage. This indicates that *-lseyla*, formerly used as either a connective or a sentence-ender, has lost its sentence-ender function in PDK.

2.4 Apprehensionals from nominalizers

2.4.1 *-cimolla* and *-cimoluni*

The source constructions of *-cimolla* and *-cimoluni* are *-ci-molu-a* and *-ci-molu-ni*. Both involve the negative-polarity nominalizer *-ci* and the lexical verb *molu* ‘not know’, which are combined with the sentence-ender/connective *-a* and the causal connective *-ni*, respectively, thus yielding the compositional meanings of ‘(I) do not know if ...’ (with the sentence-ender *-a*), ‘as (I) do not know if ...’ (with the connective *-a*), or ‘because (I) do not know if ...’ (with the causal connective *-ni*).

The forerunner constructions of these apprehensionals were first attested in the early 20th century data. The PDK example of *-cimolla* in (12) is taken from a drama.

- (12) [The speaker, tempted to check out the room of a man toward whom she has amorous feelings, now in fear of being caught, speaks to herself:]
Mos ha-y na-l cincca isanghan yeca-lu po-l-cimolla.
cannot do-END I-ACC truly weird woman-as see-FUT-APPR
‘(I) just can’t. I’m afraid (if I get caught) he might think I am a weird woman.’ (2010 Drama *Kayinuy chwihyang* Episode #9)

In PDK, the apprehensionals in this group are used at a high frequency. The apprehensional *-cimolla* is more frequently used as a sentence-ender than as a connective, and signals that the speaker fears the undesirable situation being described by the proposition, whereas *-cimoluni* is more common as a connective, sometimes occurring in fortified variant forms, *-citomolla*, *-cimolunikka*, or

-citomolunikka. A corpus search returns 5,982 hits of *-cimolla/-cimoluni* forms (including their fortified forms), of which 2,085 (34.9%) carry the apprehensional function, i.e., marking a level of apprehension in undesirable events. In other instances, they simply denote uncertainty. About 719 tokens (34.51%) of all apprehensional uses are instances of the connective function and 1,366 tokens (65.5%), of the sentence-ender function. It is also noteworthy that despite their functional similarity, their distributional patterns are quite dissimilar, i.e., for *-cimolla* it is nearly even between the connective and sentence-ender functions (49.2% vs. 50.8%), *-citomolla* occurs predominantly in the sentence-ender function (98.0%), *-cimoluni(kka)* is more specialized in the connective function (74.2%), and *-citomoluni(kka)* is found only in the connective function (100%).

2.4.2 *-cimoshakey* and *-cimoshatolok*

The source constructions of *-cimoshakey* and *-cimoshatolok* are *-ci-mos-ha-key* and *-ci-mos-ha-tolok*, which involve the negative polarity nominalizer *-ci*, negation marker *mos* ‘cannot’, and the light verb *ha-* ‘say/do’ in common, and the mode-marking adverbializer *-key* ‘in the mode of’ or the purpose-marking particle *-tolok*, respectively, thus yielding the compositional meanings of ‘in the mode in which x cannot’ (with the mode-marker *-key*), or ‘in order that x cannot’ (with the purpose-marker *-tolok*).¹² It is noteworthy in this context that *-ci-mos-ha-*, before it developed into the apprehensionals *-cimoshakey* and *-cimoshatolok*, was commonly used in negative causative constructions to mean ‘make someone unable to do x, make someone not do x’ (but not associated with apprehension), e.g., ‘make your daughter not make friends with the lowly’, ‘make someone unable to understand’, etc. In historical texts, the causer and causee in negative causative constructions are (nearly) invariably humans. These semantic features seem to have been inherited by the apprehensionals at the time of their development and then bleached as the grammaticalization processes continue (see (13) below). The conceptual motivation of these forms developing into apprehensionals is intuitively straightforward, as they carry the meaning of negative purpose, corresponding to the English apprehensional *so that ... not*.

¹²Incidentally, as an anonymous reviewer notes, in addition to negative sentences involving the nominalizer *-ci* (long-form negatives), Korean has *-ci-less* negatives (short-form negatives). Connectives from short-form negatives with the mode- or purpose-marker also carry the meaning of a situation to be avoided, e.g. *an o-key* [not come-MODE] ‘so that (he) not come’ or *mos o-tolok* [not come-PURP] ‘so that (he) not be able to come’. However, unlike those addressed in this paper, these forms lack syntagmatic cohesion by virtue of discontinuous configuration, as [NEG V-MODE/PURP], in which the verb occurs between the two grammatical markers. These peripheral apprehensionals merit future research.

The forerunner constructions of these apprehensionals were first attested in the 17th century data, with the prohibitive-apprehensive function. In PDK, these apprehensionals have not advanced much in grammaticalization morphosyntactically. However, they departed from the pure syntactic construction with a compositional meaning in that in PDK they are no longer constrained by agentivity of the implicit imposer, or by a presupposed active action of the sentential subject. The distinction between the earlier and later apprehensional connectives in this respect is illustrated in (13), in which the older (13a) involves a human subject ('lowly servants') capable of an action ('raise their head to see') whereas the newer (13b) involves an inanimate subject ('wind') incapable of voluntary action (see §3.1 for more discussion).

- (13) a. *Chyenchyul-i wulelepo-cimoshakey myengpwun-ul emhi*
 lowly.servant-NOM look.up-APPR order-ACC sternly
ha-si-ko ...
 do-HON-and
 '(You) give orders in a stern manner so that the lowly servants would not (dare to) raise their head to see you, and ...' (18xx *Hancwunglok* #84)
- b. *Palam-i tuleo-cimoshakey tantanhi kkun-to mwukk-ecwu-mye ...*
 wind-NOM enter-APPR tightly string-also tie-BEN-as
 'As (he) tightly ties the string of the jacket hood so that the wind may not come into the jacket through the hood opening, ...' (2009, Drama *Khulisumasuey nwuni olkkayo* Episode #5)

Example (14) illustrates the apprehensional *-cimoshakey* in a Modern Korean text.

- (14) *Sanyangkwn-i cumsayng-ul cap-nuntay ku cumsayng*
 hunter-NOM animal-ACC capture-while that animal
po-cimoshakey mom-ul swumky-e.
 see-APPR body-ACC hide-and
 'When a hunter is capturing an animal, (he) hides himself so that it cannot see (him).' (1903 *Sinhakwelpo* 3)

In PDK, *-cimoshakey* is moderately frequent, but *-cimoshatolok*, only occurring in literary texts, is not frequently used as an apprehensional. A corpus search returns 533 instances of *-cimoshakey*, of which 446 tokens (83.7%) are instances of the apprehensional connective usage. Other tokens are instances of manner

modifier for stative verbs, many of which are idiomatized or lexicalized (e.g., *cemcanh-cimoshakey* [be.mannerly-*cimoshakey*] ‘in an unmannerly way’). On the other hand, the search returns only 96 instances of *-cimoshatolok*, all of which are of the apprehensional connective usage.

2.4.3 *-cianhkey* and *-cianhtolok*

The source constructions of *-cianhkey* and *-cianhtolok* are *-ci-ani-ha-key* and *-ci-ani-ha-tolok*, which all have the negative-polarity nominalizer *-ci*, negation marker *ani* ‘not’, and the light verb *ha-* ‘say/do’ in common, and in addition involve the mode-marking adverbializer *-key* ‘in the mode of’ or the purpose-marking particle *-tolok*, respectively, thus yielding the compositional meanings of ‘in the mode in which x does not’ (with the mode-marker *-key*), or ‘in order that x does not’ (with the purpose-marker *-tolok*). It is clear that these are very close relatives of the apprehensionals *-cimoshakey* and *-cimoshatolok*, discussed above, the only difference being *mos* ‘cannot’ vs. *anh* ‘do not’, the former involving the semantic feature of inability. The conceptual motivation of these forms developing into apprehensionals is also intuitively straightforward, as they carry the meaning of negative purpose, corresponding to the English apprehensional *lest*.

The forerunner constructions were first attested in the 18th century data, with a weaker preventive-apprehensional function, i.e., ‘so as one does not’ (as compared to *-cimoshakey* ‘so as one cannot’). Just as is the case with their relatives, in PDK, these apprehensionals have not advanced much in grammaticalization morphosyntactically, but departed from the pure syntactic construction with a compositional meaning in that in PDK they are no longer constrained by agentivity of the implicit imposer, or presupposed active action of the sentential subject (cf. §2.4; see §3.1 for more discussion). (15) is taken from a Modern Korean source:

- (15) (The Jews to Prophet Samuel)
Wuli-lulwihaya cwuk-cianihakey ne-uy cwu hananim-skuy
 we-for die-APPR you-GEN lord God-to
kuytoha-ap-bose.
 pray-HON-OPT
 ‘[Since we have done wrong] please pray for us to your Lord God lest we die (for our sins).’ (1903 *Sinhakwelpo*)

In PDK, *-cianhkey* is highly frequent and *-cianhtolok* is moderately frequent. A corpus search returns 4,647 hits of *-cianhkey*, of which about 2,983 tokens

(64.2%) are in apprehensional use. Non-apprehensional uses involve, in parallel with *-cimoshakey* discussed above, the form occurring with stative verbs modifying them with respect to manner (e.g., *namcatap-cianhkey* [be.manly-*cianhkey*] ‘in an unmanly manner; in an ungentlemanly manner’). The instances of the apprehensional use are all in the connective function, even though some of the attested examples are inversions with the overtone of after-thought addition. The search also returns 538 hits of *-cianhtolok*, all of which are instances of the apprehensional connective use.

3 Discussion on theoretical issues

3.1 Grammaticalization parameters

Since the development of apprehensionals from compositional source constructions is an instance of grammaticalization, discussion of the developmental processes with respect to grammaticalization parameters is in order. Among a number of proposed principles and hypotheses, we adopt the mechanisms presented in Heine & Kuteva (2002: 2), i.e. desemanticization, extension, decategorialization, and erosion (for general exposition of each parameter, see Heine & Kuteva 2007: 35–44).

3.1.1 Desemanticization

Also commonly called “semantic bleaching” in figurative terms, desemanticization refers to the loss of meaning content. As briefly alluded to in the preceding exposition, lexical items have undergone desemanticization, thus creating a “divergence” (Hopper 1991) between the lexical item in the source construction and the grammatical item as a component of an apprehensional. For instance, there are six lexical items that participate in the construction of an apprehensional, namely, *po-* ‘see’ in *-lkkapwa*, *mwusep-* ‘be fearful’ in *-lkkamwusewe*, *molu-* ‘not know’ in *-lkkamolla*, *-cimolla*, and *-cimoluni*, *siph-* ‘think, inclined to think’ in *-lkkasiphe*, *sa* ‘occasion’ in *-lseyla*, and *ha-* ‘do/say’ in *-cimoshakey*, *-cimoshatolok*, *-cianhkey*, and *-cianhtolok* (note that the light verb *ha-* ‘do/say’ in *-cimoshakey* etc. is already only marginally lexical). These lexical items have undergone semantic bleaching to variable degrees as shown in Table 2, which lists the forms of the source lexemes, their glosses and lexical core meanings, and their meanings in the apprehensionals.

Semantic bleaching is not restricted to lexical items only. The functions associated with grammatical forms have also become weakened. For instance, the

Table 2: Lexical sources of Korean apprehensionals

<i>po-</i>	‘see’	visual perception	experience, be in
<i>mwusep-</i>	‘be fearful’	feeling of fear	be concerned
<i>molu-</i>	‘not know’	absence of knowledge	uncertainty
<i>siph-</i>	‘think’	cogitation	possibility
<i>ha-</i>	‘do/say’	action, locution	be
<i>sA</i>	‘occasion’	temporal event	be in

sentence-ender *-kka*, whose characteristic function is to mark an interrogative mood, does not carry this speech act meaning of posing a question when it is part of the apprehensional forms.¹³ It simply signals indeterminacy (thus uncertainty) associated with a question.

3.1.2 Extension

Also known as context generalization, extension refers to the use of a form in new contexts. The apprehensional source constructions, before their grammaticalization, retained the semantic and grammatical functions associated with each component of the source construction. This semantic-functional retention in the constructions may constrain their usage (cf. “persistence”; Hopper 1991).

However, when the constructions undergo desemanticization, as illustrated above, the constraints on their usage may be loosened, whereby the grammaticalizing forms are enabled to be used in new contexts. For instance, when the verb *po-* ‘see’ is used in the source construction of the apprehensional *-lkkapwa*, the usage is evidently restricted to a context in which the sentential subject actually visually perceives an object. When the construction is fully grammaticalized as an apprehensional, the form is not restricted to such situations. Rather, the event being described is typically *irrealis*, e.g., an imagined war in the future (as in (1b)), the possibility of having one’s silk dress stained (as in (5)), etc.

Furthermore, in §2.4, as we discussed the cases of *-cimoshakey*, *-cimoshatolok*, *-cianhkey*, and *-cianhtolok*, we noted that the negative purpose meaning of these source forms is largely retained even after grammaticalization, but that the fully grammaticalized apprehensionals are no longer constrained by the requirements

¹³The complete loss of the interrogative function in the interrogative sentence-ender *-kka* is only observable with apprehensionals. Rhetorical questions lack illocutionary force but are still perceived as questions by language users.

of agentivity of the implicit imposer, or a potential active action of the sentential subject. This can be illustrated with the grammaticalization of *-cimoshatolok*. When *-cimoshatolok* is used while the source meanings of the constructional components are fully operative, i.e., of the nominalizer (*-ci*), ‘cannot’ (*mos*), ‘do’ (*ha-*), and ‘for the purpose of’ (*-tolok*), there are restrictions imposed by these forms. For instance, the expression *o-cimoshatolok* with the verb *o-* ‘come’ would mean ‘in order to prevent (someone) from coming’, involving the purposive action of prevention on the part of the imposer (associated with the purposive *-tolok*), and the potential, but foiled, action (i.e., ‘coming’) of a volitive agent on the part of the sentential subject (associated with *mos ha-* ‘cannot do’). However, the fully grammaticalized apprehensional *-cimoshatolok* can be used in contexts not constrained by such requirements, as in ‘lest the food be exposed to the direct sunlight’ (from lit. ‘in order to prevent the direct sunlight from reaching the food’), etc. (see (13) above for similar examples).

3.1.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Grammaticalization of apprehensionals involves decategorialization. Since decategorialization typically involves, but is not restricted to, the primary categories such as nouns and verbs, we will look at such items as listed in Table 2 above, i.e. *po-* ‘see’, *mwusep-* ‘be fearful’, *molu-* ‘not know’, *siph-* ‘think’, and *sa* ‘occasion’. Among the characteristic properties of the verbal category are modifiability by adverbs, inflection by tense-aspect-modality modulation, etc. and those of the nominal category are modifiability by adjectives, pluralizability, etc. However, all the lexical forms involved in the grammaticalized apprehensionals have lost their primary-category properties, as in, for example, **-lkkacaseyhipwa* (*caseyhi* ‘carefully’ modifying *po-* ‘see’), **-lkkamaywumwusewe* (*maywu* ‘very’ modifying *mwusep-* ‘be fearful’), **-lkkacenhymolla* (*cenhye* ‘(not) at all’ modifying *molu-* ‘not know’), **-lkkakiphhisiphe* (*kipphi* ‘profoundly’ modifying *siph-* ‘think’), **-lsAtulila* (*-tul* plural marker for *sa* ‘occasion’), etc. The decategorialization evidenced by the ungrammaticality of modified forms applies to all the items listed in Table 2.

3.1.4 Erosion

Also known as phonetic reduction, erosion refers to the loss in phonetic substance. Korean has a large number of grammatical markers that have undergone extensive or drastic erosion, partly due to simplification of articulatory gestures

as well as the language's agglutinating nature, e.g., *-la-ko-ha-nun-kes-un* [END-CONN-say-ADN-thing-TOP] > *-lan* [TOP] (Rhee 2011: 764). However, this parameter is not prominent in the development of apprehensionals in Korean. As shown in Table 1 in §2.1, the apprehensionals and their source constructions do not show any meaningful degrees of phonetic erosion. The only notable reduction occurs with the apprehensionals *-cianhkey* and *-cianhtolok*, in which *ani-ha-* 'not-do' changed to *anh-*. However, this change is not specific to these apprehensionals only; rather it applies to all instances of such a string in Korean.

One aspect relevant to this relatively lesser degree of phonetic reduction is that it may have to do with the fact that the grammatical notion of apprehension in many of the apprehensionals discussed above is not (yet) fully entrenched. In other words, many of them have functional flexibility and can be used in non-apprehensional function (note, for example, the tentative intention function in *-lkkapwa*, *-lkkasiphe*, etc. in §2.2). Among the apprehensionals that do not exhibit such functional flexibility are *-lla* and *-lseyla*. This situation is captured by the 'parallel reduction hypothesis' (Bybee et al. 1994: 19–21), according to which phonological reduction and semantic bleaching occur in tandem.¹⁴ Another form that does not exhibit functional flexibility, i.e., one that is used only as an apprehensional marker, is *-lkkamwusewe*, which has not undergone much phonetic reduction. Obviously, the functional specialization of *-lkkamwusewe* as an apprehensional is due to the semantics of its source, involving the verb *mwusep-* 'be fearful'. Even so, the semantics of the source lexeme is not intact but has undergone bleaching from 'genuine fear' to 'concern' in the course of grammaticalization (see §2.2 above).

In this context there is one important aspect of functional change that merits our attention, i.e., pragmatic strengthening or conventionalization of pragmatic inference. In the cases discussed above, the source structures did not involve lexemes explicitly denoting 'fear', except for *-lkkamwusewe* from *mwusep-* 'be fearful'. They originated from constructions involving perceptual or cognitive verbs denoting 'see', 'not know', 'think', etc. (see §2.1 above). The apprehensional meanings that emerged from grammaticalization are the results of semanticization of the pragmatic inferences from the source constructions that involve such perceptual or cognitive verbs as well as the grammatical markers of indeterminacy, e.g.,

¹⁴The relationship between formal change and conceptual change is a controversial issue. Some researchers assert that formal change precedes conceptual change (e.g., Lightfoot 1989, 1991), others suggest the reverse (e.g., Sapir 1921, Givón 1975, Heine et al. 1991, Heine 1993), and still others suggest that the two cooccur (e.g., Hopper & Traugott 2003[1993], Bybee et al. 1994). It is difficult to establish a definite order, but it is evident that they interact with each other.

futurity, prospective, mode, purpose, and question. When these notions are combined, the source constructions have stronger potential to invite the inference of potential harm, which eventually becomes conventionalized (see §3.3 below for discussion on the human propensity for associating uncertainty with negative emotion, i.e., fear). Thus, the acquisition of the apprehensional meaning proceeds from a more or less neutral to a negative meaning, i.e., apprehension. This points to the fact that grammaticalization cannot be generalized as a uniformly reductive process of “erosion” or “bleaching”. Rather, there is reduction, largely with respect to phonetic substance, but there is also strengthening, largely with respect to meaning or function (cf. “the loss-and-gain model” (Heine et al. 1991: 109–111)). The role of pragmatic inference is discussed in more detail in §3.4.

3.2 Multiple layers and specialization

As is shown in the preceding exposition, there are multiple apprehensionals that emerged in the history of Korean, which have been grammaticalized to varying degrees from diverse constructions, thus forming multiple layers in contemporary Korean. A historical survey shows that the earliest forms are *-lseyla* and *-lkka*, used in Late Middle Korean. Even though *-lseyla* remains infrequent in PDK, *-lkka* is itself a moderately frequent apprehensional and furthermore has served as the basis of other future-and-question-based apprehensionals in the course of history (see §2.2), creating multiple branches of “divergence” (Hopper 1991) or “split” (Heine & Reh 1984). The developmental patterns are shown in Figure 1, in which the arbitrary groupings of “infrequent” (1–10 pmw), “moderately frequent” (11–50 pmw), and “highly frequent” (>50 pmw) are used and the thickness of the line represents the level of frequency.

When multiple forms carry an identical or similar function, there occurs functional tension among them and certain forms may gain functional primacy over others, pushing other competitors outside the functional domain or into other genre- or register-specific usage within the domain. This process has been termed “specialization” (Hopper 1991). From a statistical analysis of the frequency in the Modern Korean corpus, the primary apprehensional is *-lkkapwa*, with 276 occurrences per million words. It is followed by *-cianhkey* (123 pmw) and *-cimolla* (55 pmw). The relative functional strengths of the apprehensionals in PDK are graphically presented in Figure 2.

From Figure 1 and Figure 2, one notable aspect of the divergence and specialization patterns is that the most frequent apprehensionals in PDK are not those with the longest history of grammaticalization. Conversely, those with the shortest history are not the least frequent in PDK. For instance, the most frequently

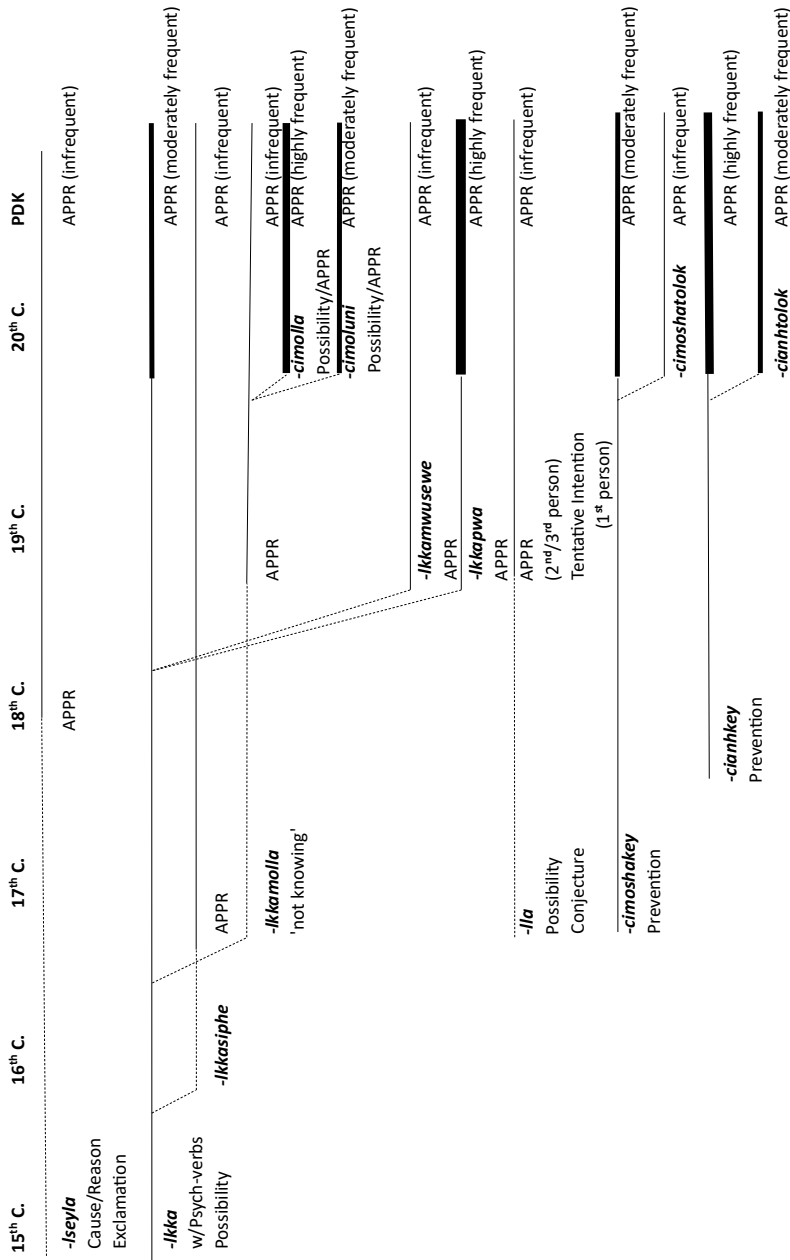


Figure 1: Historical development of apprehensionals

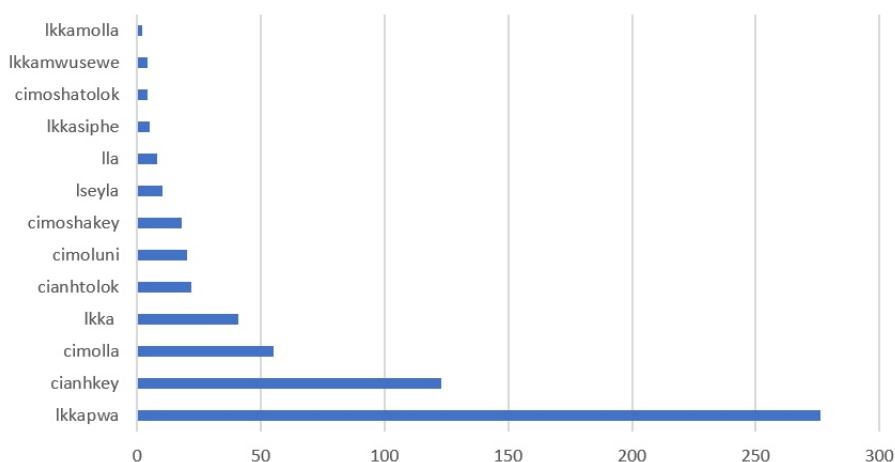


Figure 2: Frequency of apprehensionals in PDK (Drama-Movies corpus)

used apprehensionals such as *-lkkapwa*, *-cianhkey*, and *-cimolla* came to be used as apprehensionals only from the 19th, 18th, and 20th century onwards, respectively, whereas the least frequently used apprehensionals such as *-lkkamolla*, *-lkkamwusewe*, and *-cimoshatok* developed into apprehensionals from the 17th, 19th, and 17th century, respectively.

Also important in this context is the notion of the degree of grammaticalization. Even though instances of grammaticalization are often discussed as of variable degrees, e.g. “highly” grammaticalized, “weakly” grammaticalized, etc., there is no established method to measure the degree of grammaticalization. We may use the four parameters discussed above, i.e., desemanticization, extension, decategorialization and erosion, to examine how much change has occurred along these parameters. Alternatively, Lehmann’s (2015[1982]) six parameters may serve as a diagnostic, i.e., the degree of attrition (reduction of semantic content and/or phonetic substance), paradigmaticization (decrease of the paradigm size), obligatorification (use increasingly becoming obligatory), condensation (narrowing of structural scope), coalescence (increase of bondedness), and fixation (reduction of syntagmatic variability). Either way, however, since the parameters operate at different domains of grammar, such as phonology, morphology, semantics, and syntax, and since the source concepts may already be grammaticalized to some degree, different levels of change cannot be quantified for direct comparison in determining relative degrees of grammaticalization across parameters. These practical problems notwithstanding, we can hypothesize, fol-

lowing Hopper & Traugott (2003[1993]: 106), that “the more frequently a form occurs in texts, the more grammatical it is assumed to be” and that textual frequency may represent different degrees of grammaticalization of multiple forms in the same or similar functional paradigm (see, however, Heine 1992 for discussion of exceptions as well as Heine & Kuteva 2007: 38–39 for the lack of a clear correlation between frequency and grammaticalization).

From this perspective, the state of affairs of the Korean apprehensionals suggests that *-lkkapwa* is the most grammaticalized and *-lkkamolla* is the least grammaticalized, and further that the historical depth and the degree of grammaticalization are not coextensive.

3.3 The role of source concepts

The role of source semantics in grammaticalization is crucial because grammaticalization is basically a conceptually motivated process (Heine 1986, 1997, Heine & Claudi 1986, Heine et al. 1991, Kuteva 2001). We noted in §2.1 that there is a certain limited number of concepts that are involved in the source constructions, i.e. lexical verbs of perception and cognition such as ‘see’, ‘be fearful’, ‘not know’, and ‘think’ and the grammatical concepts of futurity, question, mode, and purpose. The relation between the source concepts and the resultant grammatical functions has been noted by grammaticalization researchers (cf. the source determination hypothesis by Bybee et al. (1994)), and we are going to consider the development of apprehensionals in this light.

The contribution of cognitive verbs such as ‘fear’, ‘not know’, and ‘(be inclined to) think’ and the perception verb ‘see’ to the development of apprehension is obvious, since apprehension is an inherently cognitive/perceptual state of affairs (cf. “apprehensional-epistemic” in Lichtenberk (1995)). As we noted in §2.2, the oldest predecessor construction of apprehensional *-lkk* in Late Middle Korean was followed by verbs of cognition, e.g. ‘worry’, ‘doubt’, ‘lament’, etc. (see §2.2 for a more extensive list), while *-lkk* was only a combination of the future marker (*-l-*) and the interrogative sentence-ender (*-kka*). When the internal cohesion in the collocation increased, the collocational constructions became formally unverbated and conceptually integrated into the set of apprehensionals, thus leading to the emergence of multiple apprehensionals, e.g. *-lkkasiphe* (with ‘think’), *-lkkamolla* (with ‘not know’), *-lkkamwusewe* (with ‘be fearful of’), and *-lkkapwa* (with ‘see’).¹⁵

¹⁵Conceptual integration cannot be easily proven with quantifiable criteria. In Korean, however, one such indication appears in the way people write. These apprehensionals are structurally

It is interesting to note that in the development of apprehensionals from *-lkka*, the ‘fear’-semantics of the co-occurring verbs or propositions denoting undesirable events seems to have been absorbed by *-lkka*, a process whereby *-lkka* itself became an apprehensional around the 18th century. This is a plausible scenario, a process that Bybee et al. (1994) labeled ‘absorption of contextual meaning’ and included among the possible grammaticalization mechanisms (see Bybee et al. 1994: 230–236 for illustration of modal meanings). Similarly, Rhee (2017) shows that the semantics of many sentence-final particles that developed from insubordination are in fact the inferred meanings of the imagined reconstruction of an elided main clause, instances of borrowed semanticization from neighboring forms. A caveat is that the process of borrowed semanticization or absorption has not happened in full, and as a result there are non-apprehensional usages of *-lkka* in PDK (see §2.2).

Another important aspect of the contribution of source semantics is that not only the lexical semantics but also grammatical concepts associated with the source constructions are important contributors. It is evident that futurity plays an important role in the development of apprehensionals, as shown by the fact that seven out of thirteen developed from constructions containing the future tense marker *-l-*. The semantics of the particles for mode (*-key*) and purpose (*-tolok*), each involved in two apprehensionals, is also inherently futuristic, in the sense that the event marked by them is intended by the (implied) sentential subject but has not yet been realized. Interrogative marking also plays an important role.¹⁶ Five of the apprehensional markers have the question marker *-kka* as a component. The grammatical notions of futurity and interrogative are significant because they bring forth the notions of uncertainty and indeterminateness, which, in turn, engender the notion of apprehension. It is true that the source of apprehension does not have to be uncertainty, considering that we may become particularly apprehensive when we are positive that something bad is going to happen.¹⁷ However, it is also true that the uncertainty-anxiety relationship has long been recognized in human psychology and neuroscience. For instance, it has been noted that the human brain is an “anticipation machine” (Gilbert 2006,

compositional and thus need to be written with inter-lexical spacing according to orthographic rules, but they are often written without spaces at the word boundary in informal writing (e.g., *-lkkasiphe* instead of *-lkka siphe*; *-lkkamolla* instead of *-lkka molla*, etc.). Incidentally, disappearance of interlexical spacing is commonly observed in the grammaticalization of multi-word constructions such as auxiliary verbs, complex postpositions, sentence-final particles, etc.

¹⁶The connective *-key*, also often referred to as a converb marker, is polyfunctional but its primary function is marking the mode of the action or state denoted by the verb (Rhee 1996). For diverse functions of *-key* in grammaticalized constructions, see Kim (2011).

¹⁷We thank an anonymous reviewer who brought this insightful point to our attention.

as cited in Grupe & Nitschke 2013: 488), and that people have a general propensity toward negative affective responses to uncertainty (Anderson et al. 2019, see also Carleton 2016a,b).

3.4 Pragmatic inference, reinterpretation, and subjectification

It is widely recognized that grammaticalization often results from pragmatic inference (Traugott 1989, Hopper & Traugott 2003[1993], Slobin 1994, Bybee et al. 1994, Fischer 2008, and numerous others). The process can be illustrated with the apprehensional connective *-lkkapwa*, developed from *-l-kka-po-a* [FUT-Q-see-CONN], exemplified in (1b), with modification:

(16) Connective *-lkkapwa* ‘for fear of’

- a. *Ku-nun cencayng-i na-lkkapwa cam-ul mos ca-n-ta.*
 he-TOP war-NOM come.out-APPR sleep-ACC cannot sleep-PRS-DECL
 ‘He cannot sleep for fear of a war.’
- b. *Ku-nun cencayng-i na-l-kka pw-a cam-ul mos*
 he-TOP war-NOM come.out-FUT-Q see-CONN sleep-ACC cannot
ca-n-ta.
 sleep-PRS-DECL
 Lit. ‘As he sees/saw (it), (saying,) “Will a war break out?”, he cannot sleep.’

Before the poly-morphemic *-lkkapwa* was fully grammaticalized, the meaning of the example (16a), based on the components constituting *-lkkapwa*, would have been (16b), ‘As he sees/saw (it), (saying,) “Will a war break out?”, he cannot sleep.’ This interpretation, faithful to the meanings of the source components, makes reference to ‘his seeing (or having seen) something’. However, since the war has not (yet) broken out, actual occurrence of visual perception of the war is not likely. Probable events may be only hearing rumors about a war or sensing the possibility in one way or another. Therefore, making reference to seeing (by using the verb *po-* ‘see’) may be interpreted by the hearer as either a hyperbole or a metaphor of expressing a cognitive state (i.e. knowledge) by means of bodily perception (i.e. seeing), a crosslinguistically common instance of metaphorization (see Lakoff & Johnson 1980, Sweetser 1990).

Also, it is not clear if the referent of *ku-nun* ‘he’ in (16a) really uttered the question “Will a war break out?” In fact, it is not likely that the question was actually asked by him, because *-l-kka* as a regular future interrogative marker (not as an apprehensional) is typically a monologic sentence-ender (a type of “audience-blind”

form, Rhee & Koo 2017: 105).¹⁸ In other words, the speaker imagines that the sentential subject of an apprehensional sentence asks himself or herself if something will happen. Thus, the question exists only in the mind of the speaker. In this process, the sentence-ender of the question in the future tense is pragmatically reinterpreted as a connective of a clause that marks the cause of apprehension. This type of figurative hyperbole is a common expressive strategy in Korean lexicalization and grammaticalization, observed by Rhee (2009), who labels it a “through a borrowed mouth” phenomenon.¹⁹ What is noteworthy is that all this expressive enrichment involves pragmatic inference, i.e., from “a question is a speech act involving the speaker’s uncertainty” to “uncertainty about the future generates fear” and further to “a question about the future marks fear”.

Pragmatic inference may well – but does not have to – lead to subjectification, a process whereby “meanings based in the external described situation [become] meanings based in the internal (evaluative/perceptual/cognitive) situation”, or “meanings tend to become increasingly situated in the speaker’s subjective belief-state/attitude toward the situation” (Traugott & König 1991: 208–209), a frequent semantic concomitant in grammaticalization (Traugott 2003). In this light, the semantic/functional change through pragmatic inference described above is a paradigm example of subjectification, i.e., the change from ‘as one sees something and asks “Will x occur?”’ to ‘for fear of x’ is a shift from a description of the world (i.e., the external described situation) to one of a mental world (i.e., the meaning situated in the speaker’s subjective attitude).

Similarly, the development of apprehension from uncertainty and indeterminateness is a clear instance of subjectification. Uncertainty is a cognitive state often resulting from the indeterminateness of a situation, whereas apprehension is more subjective by virtue of being an affective state. Even when apprehensionals do not mark fear but potential undesirability, the evaluative judgment of a future event involves subjectivity.²⁰ Similarly, the development of the verb *molu*- ‘not know’ to the apprehensionals *-lkkamolla*, *-cimolla*, and *-cimoluni* is

¹⁸For this characteristic of the sentence-ender *-lkkka*, the clause headed by *-lkkapwa*, embedding a question thus marked, represents a situation in which the sentential subject is asking a monologic question to himself/herself while looking at an event or state.

¹⁹Rhee (2009) lists descriptive adverbials that originated from quotation constructions, e.g., *cwuk-keyss-ta-ko* ‘saying “I will die”’ to *cwukkeyssstako* ‘desperately’, *michyess-ta-ko* ‘saying “I am insane”’ to *michyesstako* ‘nonsensically’, etc. The quotations “I will die”, “I am insane” did not originate from the person being described but from the imagination of the observer.

²⁰It is noteworthy in this context that Kim (2021) notes in her analysis of *-ltheyntey* from a connective (e.g., ‘while P will occur’) to a final particle (e.g., ‘It is regrettable that P’) that the grammaticalization involves semantic extension into wishing, regretting, worrying, etc., in which conjecture and context play important roles.

an instance of subjectification, along the series of changes as ‘have no knowledge’ > ‘be unsure’ > ‘be uncertain’ > ‘be worrisome’ > ‘be apprehensive’. The series of changes along the conceptual continuum strongly suggests developmental directionality from epistemic possibility to apprehension (Dobrushina 2006, Pakendorf & Schalley 2007, among others).

When pragmatic inferences occur, often involving subjectification, there also occur morphosyntactic reanalysis and functional reinterpretation, i.e., a polymorphemic construction with compositional meaning being reanalyzed as a single form with grammatical meaning. Certain types of morphosyntactic change can result in much more drastic consequences. For instance, we noted in §2.1 that the resultant grammatical category (i.e. connective or sentence-ender) of most instances of apprehensionals is identical with the grammatical category of the final element in the source construction and that there are three exceptional cases, i.e. *-lkka*, *-lla*, and *-lseyla* ((a), (f), and (g) in Table 1). They all end with a sentence-ender: *-lkka* with an interrogative sentence-ender *-kka*, and *-lla* and *-lseyla* with a declarative sentence-ender with some nuance of exclamative, *-la*. However, *-lkka* is now an apprehensional connective, not an apprehensional sentence-ender, and *-lla* and *-lseyla* end with a sentence-ender *-la* but now as apprehensionals they also function as connectives. This state of affairs warrants some discussion.

The source construction of *-lkka*, as noted in §2.2, is an interrogative sentence in the form of direct quotation, i.e. an embedded question. However, when the direct quote provides a cause/reason as the background of the main clause, the form at the boundary of embedding, i.e. the end of the embedded sentence, *-lkka*, is functionally reinterpreted as a causal connective (linking the state or event feared by the speaker or the sentential subject to its consequence). Let us look at the following example (17):

- (17) a. *Cencaying-i na-l-kka* *cam-i* *an o-n-ta*.
 war-NOM come.out-FUT-Q.END sleep-NOM not come-PRS-DECL
 ‘Will a war break out? (I) cannot go to sleep.’
- b. *Cencaying-i na-l-kka* *cam-i* *an o-n-ta*.
 war-NOM come.out-APPR.CONN sleep-NOM not come-PRS-DECL
 ‘(I) cannot sleep for fear of a war (breaking out).’

The change from (17a) to (17b) clearly shows the inter-clausal morphosyntactic reanalysis of regarding the sentence-final *-l-kka* as a clause-final connective and functional reinterpretation of regarding the marker of a question with respect

to the future as a marker of apprehension, thus from a sentence-ender to a connective. As a matter of fact, this type of change is pervasive in the grammaticalization of connectives in Korean, which involves “self-quotative constructions” (SQC) (Koo & Rhee 2019: 256), *-lkka* being one of them. Example (18) illustrates the SQC connective *-na*, whose original function is marking an interrogative sentence, just like the apprehensional *-lkka* being discussed here:

- (18) a. Connective (causal) *-na*
Amwu-to eps-na coyongha-ta.
 anyone-even not.exist-CONN be.quiet-DECL
 ‘It’s quiet, perhaps because there’s nobody around.’
 b. Source construction *-na* ‘-Q’
Amwu-to eps-na coyongha-ta.
 anyone-even not.exist-Q be.quiet-DECL
 Lit. “‘Is nobody here?’ it is quiet.” (Koo & Rhee 2019: 267)

The source-construction example (18b) consists of two sentences: one monologic question and one (subject-less) declarative sentence. It is noteworthy that two juxtaposed sentences of this type can undergo coalescence (Haspelmath 2011) into a single sentence (as in (18a)). In this process, the sentence-final element (the question particle *-na* in (18b)) develops into a connective (the causality marker in (18a)). The examples in (18) show a change analogous to the one with *-lkka*, from a sentence-ender to a connective; specifically, from a question marker to a causal connective (note, however, that *-na* does not encode apprehension).

A similar pattern is also observed in the development of apprehensional connectives from apprehensional sentence-enders, such as *-lla* and *-lseyla*. The apprehensionals *-lla* and *-lseyla* end with a sentence-ender *-la* in their source constructions. Previous studies analyze them as sentence-enders (Ramstedt 1997[1939], Kim 1960, Ko 2002[1987], Jang 2002). Let us look at the example (10) illustrating the apprehensional *-lla*, repeated here for convenience as (19a):

- (19) a. [In preparation of a funeral]
Pang tewu-lla pwul stay-cimal-ko ko-y tuleka-lla
 room be.hot-APPR fire make-PROH-and cat-NOM enter-APPR
kwistol mak-ela.
 flue.opening block-IMP
 ‘Don’t make fire (for floor-heating) lest the room become too hot (for the corpse), and block the flue opening lest a cat enter (through it) [which is ominous according to superstitions].’ (ca. 1870
Pakhungpoka)

- b. *Pang tewu-l-la pwul stay-cimal-ko ko-y tuleka-l-la*
 room be.hot-FUT-END fire make-PROH-and cat-NOM enter-FUT-END
kwistol mak-ela.
 flue.opening block-IMP
 ‘I’m afraid the room will become too hot. Don’t make fire (for
 floor-heating), and I’m afraid a cat will enter. Block the flue opening.’

The translation (19a) of the example is based on the interpretation of *-lla* as a connective. If we follow the analyses referenced above, the example would be translated as (19b) with the structure of: [S₁ I’m afraid the room will become too hot.] [S₂ Don’t make fire (for floor-heating)], and [S₃ I’m afraid a cat will enter.] [S₄ Block the flue opening.], containing four finite sentences, with S2 and S3 constituting a coordinated sentence, combined by the coordinating connective *-ko* ‘and’. Note, however, that conceptual cohesion does not exist between S2 and S3. Instead, the conceptual cohesion is found between S1 and S2, and between S3 and S4, respectively. In other words, S1 is the reason for saying S2, and S3 for saying S4. The connections between S1 and S2 and between S3 and S4 are pragmatic causality or “enablement” (cf. Sweetser 1990: 77).²¹ Therefore, the example is better analyzed as a sentence constituting two sentences in coordination, each consisting of a subordinate clause and an imperative main clause: [S[S₁[SUB.CL Reason-APPR.CONN] [MAIN.CL Command]]-COORD [S₂[SUB.CL Reason-APPR.CONN] [MAIN.CL Command]]]. In other words, the connective interpretation in (19a) is better than the sentence-ender interpretation in (19b). This situation is closely related to the crosslinguistic observation that apprehensionals are often used in the contexts of “imperative indicating the action the hearer should undertake to avoid the unpleasant outcome” and “prohibitive indicating the action the hearer should avoid” (Schultze-Berndt 2015) and that apprehensionals are linked to warning/threat and negative command (prohibitive) (Pakendorf & Schalley 2007, as cited in Schultze-Berndt 2015).

Therefore, the pragmatic motivation for attributing a causal relation between two sentences and for redefining the grammatical status of the sentences seems to have led to the reanalysis of a former sentence-ender into a connective. This type of reanalysis has occurred in the grammaticalization of former sentence-enders into connectives (as shown in (18) and (19) above, and conversely, and more frequently, in the grammaticalization of former connectives into sentence-enders through the processes of *insubordination* (Evans 2007, Evans & Watanabe

²¹In her discussion of conjunctions with an example of “What are you doing tonight, because there’s a good movie on,” Sweetser (1990: 77) interprets the function of the causal conjunction *because* as a means of justifying and enabling the question, thus the cause of speech act.

2016) and *incoordination* (Kuteva et al. 2015, 2017). In fact, pragmatic inferences that operate in insubordination and incoordination prompted the functional extension of some apprehensional connectives to the paradigm of sentence-final particles, marking tentative intention, doubt, indeterminateness, possibility, etc. (Koo & Rhee 2001, 2013, Sohn 2003, Rhee & Koo 2017).

4 Summary and conclusion

Despite the fact that Korean has a large number of apprehensionals, their developmental paths have not received any serious analysis to date. As we intended to address all apprehensionals being used in PDK, we were unable to trace all the grammaticalization trajectories of the individual forms diachronically, but we have shown that Korean apprehensionals have differential historical depths and varying degrees of specialization. They form multiple layers in the functional domain of apprehensionals in PDK.

Among the four main interrelated grammaticalization mechanisms (Heine & Kuteva 2002: 2), i.e., desemanticization, extension, decategorialization, and erosion, erosion is the least prominent one. Unlike highly grammaticalized markers which tend to be short in form, many apprehensionals are still at the periphrastic stage in morpho-syntactic terms. This suggests that apprehensional markers in Korean have not acquired the status of full-fledged apprehensionals. In fact, many of them can still be used for non-apprehensional functions. This functional split is prominent in their use as an apprehensional connective and as a non-apprehensional sentence-ender. Further, some apprehensionals originated from structurally complex constructions through strengthening of pragmatic inferences based on the semantics of the components, notably epistemic modals, whereas the ‘fear’-based apprehensionals incorporated a ‘fear’ verb and thus their apprehensional function was visible throughout their history. Multiple forms synchronically layered in the functional domain of apprehensionals have variable specializations as shown in their frequency of use.

Lexical semantics as well as the functions of grammatical forms in the source constructions contributed to the creation of the notion of apprehension. Of particular importance are (i) the perception/cognition verbs denoting ‘be fearful’, ‘not know’, ‘think’, and ‘see’ in the lexical sources, and (ii) futurity, interrogative, mode and purpose from the grammatical formants.

In the course of development of apprehensionals, pragmatic inferences, subjectification, morphosyntactic reanalysis, and functional reinterpretation are prominent driving forces. In particular, the cross-categorical flexibility between connectives and sentence-enders is largely due to the pragmatic motivations that

attribute a causal relation between two sentences and redefine the grammatical status of the sentences. These developmental patterns are not isolated instances but are commonly attested in the grammaticalization scenarios involving insubordination and incoordination in the history of Korean.

Abbreviations

ACC	accusative	NOM	nominative
ADN	adnominalizer	NOMZ	nominalizer
APPR	apprehensional	OPT	optative
BEN	benefactive	PDK	present day Korean
COMP	complementizer	PL	plural
CONJ	conjecture	PROG	progressive
CONN	connective	PROH	prohibitive
DECL	declarative	PRS	present
END	sentence-ender	PST	past
FUT	future	PURP	purposive
GEN	genitive	Q	question particle/marker
HON	honorific	RETRO	retrospective
IMP	imperative	TOP	topic
IND	indicative	VOC	vocative
MODE	mode-adverbializer		

Acknowledgements

We are thankful to the anonymous reviewers of the manuscript and to Eva Schultze-Berndt, Marine Vuillermet, and Martina Faller for their constructive comments and suggestions. This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2020S1A5A2A03043203).

References

- Anderson, Eric C., R. Nicholas Carleton, Michael Diefenbach & Paul K. J. Han. 2019. The relationship between uncertainty and affect. *Frontiers in Psychology* 10. Article nr 2504. DOI: 10.3389/fpsyg.2019.02504.
- Bybee, Joan L. & William Pagliuca. 1985. Crosslinguistic comparison and the development of grammatical meaning. In Jacek Fisiak (ed.), *Historical semantics — historical word formation*, 59–84. Berlin: Mouton.

- Bybee, Joan L., Revere D. Perkins & William Pagliuca. 1994. *The evolution of grammar: Tense, aspect, and modality in the languages of the world*. Chicago: University of Chicago Press.
- Carleton, Nicholas. 2016a. Fear of the unknown: One fear to rule them all? *Journal of Anxiety Disorders* 41. 5–21. DOI: 10.1016/j.janxdis.2016.03.011.
- Carleton, Nicholas. 2016b. Into the unknown: A review and synthesis of contemporary models involving uncertainty. *Journal of Anxiety Disorders* 39. 30–43. DOI: 10.1016/j.janxdis.2016.02.007.
- Crowley, Terry. 2000. Simplicity, complexity, emblematicity, and grammatical change. In Jeff Siegel (ed.), *Processes of language contact: Studies from Australia and the South Pacific*, 175–193. Montréal: Les Éditions Fides.
- Dobrushina, Nina R. 2006. Grammatičeskie formy i konstrukcii so značeniem opasenija i predostereženija [Grammatical forms expressing warning and apprehension]. *Voprosy jazykoznanija* 2. 28–67.
- Evans, Nicholas. 2007. Insubordination and its uses. In Irina Nikolaeva (ed.), *Finiteness: Theoretical and empirical foundations*, 366–431. Oxford: Oxford University Press.
- Evans, Nicholas & Honoré Watanabe. 2016. *Insubordination*. Amsterdam: John Benjamins.
- Fischer, Olga. 2008. On analogy as the motivation for grammaticalization. *Studies in Language* 32(2). 336–382. DOI: 10.1075/sl.32.2.04fis.
- Gilbert, Daniel T. 2006. *Stumbling on happiness*. New York: Random House.
- Givón, Talmy. 1975. Serial verbs and syntactic change. In Charles Li (ed.), *Word order and word order change*, 47–112. Amsterdam: John Benjamins.
- Grace, George W. 1997. On the changing context of Austronesian historical linguistics. In Cecilia Odé & Wim Stokhof (eds.), *Proceedings of the seventh international conference on Austronesian linguistics*, 15–32. Amsterdam: Rodopi.
- Grupe, Dan W. & Jack B. Nitschke. 2013. Uncertainty and anticipation in anxiety: An integrated neurobiological and psychological perspective. *Nature Reviews Neuroscience* 14(7). 488–501. DOI: 10.1038/nrn3524.
- Haspelmath, Martin. 1995. The converb as a cross-linguistically valid category. In Martin Haspelmath & Ekkehard König (eds.), *Converbs in cross-linguistic perspective: Structure and meaning of adverbial verb forms – Adverbial participles, gerunds*, 1–55. Berlin: Mouton de Gruyter.
- Haspelmath, Martin. 2011. The gradual coalescence into ‘words’ in grammaticalization. In Heiko Narrog & Bernd Heine (eds.), *The Oxford handbook of grammaticalization*, 342–355. Oxford: Oxford University Press.

- Heine, Bernd. 1986. *The rise of grammatical categories: Cognition and language change in Africa*. Bloomington, IN: African Studies Program, Indiana University.
- Heine, Bernd. 1992. *Some principles of grammaticalization*. An invited lecture at the Stanford/Berkeley Grammaticalization Workshop.
- Heine, Bernd. 1993. *Auxiliaries: Cognitive forces and grammaticalization*. Oxford: Oxford University Press.
- Heine, Bernd. 1997. *Cognitive foundations of grammar*. Oxford: Oxford University Press.
- Heine, Bernd & Ulrike Claudi. 1986. *On the rise of grammatical categories: Some examples from Maa*. Berlin: Dietrich Reimer.
- Heine, Bernd, Ulrike Claudi & Friederike Hünemeyer. 1991. *Grammaticalization: A conceptual framework*. Chicago: The University of Chicago Press.
- Heine, Bernd & Tania Kuteva. 2002. *World lexicon of grammaticalization*. Cambridge: Cambridge University Press.
- Heine, Bernd & Tania Kuteva. 2007. *The genesis of grammar: A reconstruction*. Oxford: Oxford University Press.
- Heine, Bernd & Mechthild Reh. 1984. *Grammaticalization and reanalysis in African languages*. Hamburg: Helmut Buske.
- Hopper, Paul J. 1991. On some principles of grammaticalization. In Elizabeth C. Traugott & Bernd Heine (eds.), *Approaches to grammaticalization*, vol. 1, 17–35. Amsterdam: John Benjamins.
- Hopper, Paul J. & Elizabeth C. Traugott. 2003[1993]. *Grammaticalization*. 2nd edn. Cambridge: Cambridge University Press.
- Jang, Yoonhee. 2002. *Cwungseykwuke congkyeleme yenkwu* [A study on the sentence-enders in Middle Korean]. Seoul: Kwukehakhoy.
- Jeong, Howan. 2003. *Hankwukeyu paltalkwa uyconmyengsa* [The development of Korean and dependent nouns]. Seoul: Ihoymwunhawa.
- Jing-Schmidt, Zhuo & Vsevolod Kapatsinski. 2012. The apprehensive: Fear as endophoric evidence and its pragmatics in English, Mandarin, and Russian. *Journal of Pragmatics* 44. 346–373. DOI: 10.1016/j.pragma.2012.01.009.
- Kim, Min-soo. 1960. *Kwuke mwunpeplon yenkwu* [A study of Korean grammar]. Seoul: Tongmungwan.
- Kim, Minju. 2010. The historical development of Korean *siph-* ‘to think’ into markers of desire, inference, and similarity. *Journal of Pragmatics* 42(4). 1000–1016. DOI: 10.1016/j.pragma.2009.08.010.
- Kim, Minju. 2011. The historical development of the Korean suffix *-key*. In Ho-Min Sohn, Haruko Minegishi Cook, William O’Grady, Leon A. Serafim & Sang

- Yee Cheon (eds.), *Japanese/Korean linguistics*, vol. 19, 435–448. Stanford: CSLI Publications.
- Kim, Minju. 2021. Development of the Korean connective *ultheyntey* into a final particle of wishing and worrying. *Journal of Pragmatics* 173. 51–65. DOI: 10.1016/j.pragma.2020.12.004.
- Kim, Tae Yeop. 2001. *Kwuke congkyelemiuy mwunpep* [Grammar of Korean sentence-final markers]. Seoul: Kwukhakcalyowen.
- Ko, Yong-Kun. 1989. *Kwuke hyengthaylon yenkwu* [Studies on Korean morphology]. Seoul: Seoul National University Press.
- Ko, Yong-Kun. 2002[1987]. *Phyocwun cwungsey kwuke mwunpeplon* [Middle Korean grammar]. Revised edition. Seoul: Jipmundang.
- Koo, Hyun Jung. 1987. Ssikkuth -a, -key, -ci, -ko-uy ssuimkwa uymi [The usage and meaning of suffixes -a, -key, -ci and -ko]. *Konkuk Emwunhak* 11–12. 167–188.
- Koo, Hyun Jung & Seongha Rhee. 2001. Grammaticalization of a sentential end marker from a conditional marker. *Discourse and Cognition* 8(1). 1–19.
- Koo, Hyun Jung & Seongha Rhee. 2013. On an emerging paradigm of sentence-final particles of discontent: A grammaticalization perspective. *Language Sciences* 37. 70–89. DOI: 10.1016/j.langsci.2012.07.002.
- Koo, Hyun Jung & Seongha Rhee. 2019. From self-talk to grammar: The emergence of multiple paradigms from self-quoted questions in Korean. *Lingue e Linguaggi* 31. 255–278. DOI: 10.1285/i22390359v31p255.
- Kuteva, Tania. 2001. *Auxiliation: An enquiry into the nature of grammaticalization*. Oxford: Oxford University Press.
- Kuteva, Tania. 2009. Grammatical categories and linguistic theory: Elaborateness in grammar. In Peter K. Austin, Oliver Bond, Monik Charette, David Nathan & Peter Sells (eds.), *Proceedings of the Language Documentation and Linguistic Theory conference 2*, 13–28. London: SOAS.
- Kuteva, Tania, Bas Aarts, Gergana Popova & Anvita Abbi. 2019. The grammar of ‘nonrealization’. *Studies in Language* 43(4). 850–895. DOI: 10.1075/sl.18044.kut.
- Kuteva, Tania, Bernd Heine, Peter Austin, Seongha Rhee, Marine Vuillermet & Domenico Niclot. 2017. *The “mirror” of insubordination*. Invited lecture, Linguistics, SOAS, University of London, June 2017. <https://www.youtube.com/watch?v=Jfq1KcoRens>.
- Kuteva, Tania, Bernd Heine, Marine Vuillermet & Domenico Niclot. 2015. *Insubordination and incoordination: pre-requisite for grammaticalization of connectives*. Paper presented at the Grammaticalization Workshop. Radboud University, Nijmegen, The Netherlands, September 28, 2015.

- Lakoff, George & Mark Johnson. 1980. *Metaphors we live by*. Chicago: University of Chicago Press.
- Lehmann, Christian. 2015[1982]. *Thoughts on grammaticalization*. 3rd edn. Berlin: Language Science Press.
- Lichtenberk, František. 1991. Semantic change and heterosemy in grammaticalization. *Language* 67(3). 475–509.
- Lichtenberk, František. 1995. Apprehensional epistemics. In Joan L. Bybee & Suzanne Fleischman (eds.), *Modality in grammar and discourse* (Typological Studies in Language 32), 293–327. Amsterdam: John Benjamins. DOI: 10.1075/tsl.32.12lic.
- Lightfoot, David. 1989. The child's trigger experience. *Brain and Behavioral Sciences* 12. 321–375.
- Lightfoot, David. 1991. *How to set parameters: Arguments from language change*. Cambridge, MA: The MIT Press.
- Pakendorf, Brigitte & Ewa Schalley. 2007. From possibility to prohibition: A rare grammaticalization pathway. *Linguistic Typology* 11(3). 515–540. DOI: 10.1515/LINGTY.2007.032.
- Park, Young-Jun. 1996. Kwuke panmal congkyelemiuy yeksaseng: -e-wa -ci-lul cwungsimulo [A historical analysis of Korean half-talk sentence-enders: With particular focus on -e and -ci]. *Korea University Emwunnoncip* 35. 97–118.
- Ramstedt, Gustav J. 1903. *Über die Konjugation des Khalkha-Mongolischen* [On the conjugation of Khalkha-Mongolian]. Helsinki: Société Finno-Ougrienne.
- Ramstedt, Gustav J. 1997[1939]. *A Korean grammar*. Helsinki: Mémoires de la Société Finno-ougrienne LXXXII.
- Rhee, Seongha. 1996. *Semantics of verbs and grammaticalization: The development in Korean from a cross-linguistic perspective*. The University of Texas at Austin. (Doctoral dissertation).
- Rhee, Seongha. 2007. Particle selection in Korean auxiliary formation. In Raúl Aranovich (ed.), *Split auxiliary systems*, 237–254. Amsterdam: John Benjamins.
- Rhee, Seongha. 2009. Through a borrowed mouth: Reported speech and subjectification in Korean. *LACUS Forum* 34. 201–210.
- Rhee, Seongha. 2011. Grammaticalization in Korean. In Bernd Heine & Heiko Narrog (eds.), *The Oxford handbook of grammaticalization*, 764–774. Oxford: Oxford University Press.
- Rhee, Seongha. 2017. *Grammar ex nihilo: a story of ellipsis*. Invited lecture at the Fall 2017 Linguistics Society of Korea Conference, Seoul National University, Nov. 25, 2017.
- Rhee, Seongha. 2019. On polygrammaticalization of see-derived auxiliaries in Korean. *Discourse and Cognition* 26(2). 91–119. DOI: 10.15718/discog.2019.26.2.91.

- Rhee, Seongha. 2020. Pseudo-hortative and the development of the discourse marker *eti poca* 'well, let's see' in Korean. *Journal of Historical Pragmatics* 21(1). 53–82. DOI: 10.1075/jhp.00036.rhe.
- Rhee, Seongha. 2021a. *Inferential evidentials and their discourse functions: A case of 'see' and 'be like' in Korean*. Paper presented at the Pragmatics of Evidentiality: Inter- and Intra-linguistic Perspectives on Pragmatic Aspects of Evidentiality Marking in Communication Conference, The University of Ca' Foscari, Venice, Italy, June 19–21, 2021.
- Rhee, Seongha. 2021b. *Seeing and thinking in inferential evidentiality: The case in Korean*. Paper presented at the Evidentiality and Modality: At the Crossroads of Grammar and Lexicon Conference, Montpellier, France, June 10–11, 2021.
- Rhee, Seongha. 2021c. The journey of thinking: From subjectivity to textuality and to intersubjectivity. In Gabi Danon (ed.), *Proceedings of IATL 2018–2019* (MIT Working Papers in Linguistics #92).
- Rhee, Seongha & Hyun Jung Koo. 2017. Audience-blind sentence-enders in Korean: A discourse-pragmatic perspective. *Journal of Pragmatics* 120. 101–121. DOI: 10.1016/j.pragma.2017.09.002.
- Rhee, Seongha & Hyun Jung Koo. 2021. On divergent paths and functions of 'background'-based discourse markers in Korean. In Alexander Haselow & Sylvie Hancil (eds.), *Studies at the grammar-discourse interface*, 77–100. Amsterdam: John Benjamins.
- Rice, Sally. 2017. Phraseology and polysynthesis. In Michael Fortescue, Marianne Mithun & Nicholas Evans (eds.), *The Oxford Handbook of Polysynthesis*, 203–214. Oxford: Oxford University Press.
- Rice, Sally. 2019. *Cognitive linguistics and the study of indigenous languages*. Plenary lecture at the 15th International Cognitive Linguistics Conference (ICLC-15): Cross-linguistic Perspectives on Cognitive Linguistics, Kwansei Gakuin University, Nishinomiya, Japan, August 6–11, 2019.
- Sapir, Edward. 1921. *Language: An introduction to the study of speech*. New York: Hartcourt, Brace & World.
- Schultze-Berndt, Eva. 2015. *The grammar of fear in Australian languages and beyond*. Seminar lecture at the Laboratoire Dynamique du Langage, CNRS, April 3, 2015.
- Simons, Gary F. & Charles D. Fennig (eds.). 2018. *Ethnologue: Languages of the world*. 21st edn. Dallas, Texas: SIL International. <http://www.ethnologue.com>.
- Slobin, Dan I. 1994. Talking perfectly: Discourse origins of the present perfect. In William Pagliuca (ed.), *Perspectives on grammaticalization*, 119–134. Amsterdam: John Benjamins.

- Sohn, Sung-Ock S. 2003. On the emergence of intersubjectivity: An analysis of the sentence-final *nikka* in Korean. *Japanese/Korean Linguistics* 12. 52–63.
- Son, Se-Mo-dol. 1994. *Cwungseykwuke pocoyongen* [Auxiliary verbs in Middle Korean]. Seoul: Hankook Publisher.
- Suh, Chung Mok. 1993. Kwuke kyengepepuy pyenchen [Historical change of the Korean honorification system]. *Institute of Korean Studies Hankwukemwun* 2. 107–145.
- Sweetser, Eve E. 1990. *From etymology to pragmatics: Metaphorical and cultural aspects of semantic structure*. Cambridge: Cambridge University Press.
- Thurston, William R. 1988. How exoteric languages build a lexicon: Esoterogeny in West New Britain. In Ray Harlow & Robin Hopper (eds.), *VICAL I: Papers from the fifth international conference on Austronesian languages*, 555–579. Auckland: Linguistic Society of New Zealand.
- Traugott, Elizabeth C. 1989. On the rise of epistemic meanings in English: An example of subjectification in semantic change. *Language* 65. 31–55.
- Traugott, Elizabeth C. 2003. From subjectification to intersubjectification. In Raymond Hickey (ed.), *Motives for language change*, 124–139. Cambridge: Cambridge University Press.
- Traugott, Elizabeth C. & Ekkehard König. 1991. The semantics-pragmatics of grammaticalization revisited. In Elizabeth C. Traugott & Bernd Heine (eds.), *Approaches to grammaticalization, Vol. 1*, 189–218. Amsterdam: John Benjamins.
- Underwood, Horace Grant. 1915[1890]. *An introduction to the Korean spoken language*. Yokohama: Kelly & Walsh.
- Wray, Alison & George W. Grace. 2007. The consequences of talking to strangers: Evolutionary corollaries of socio-cultural influences on linguistic form. *Lingua* 117(3). 543–578.

